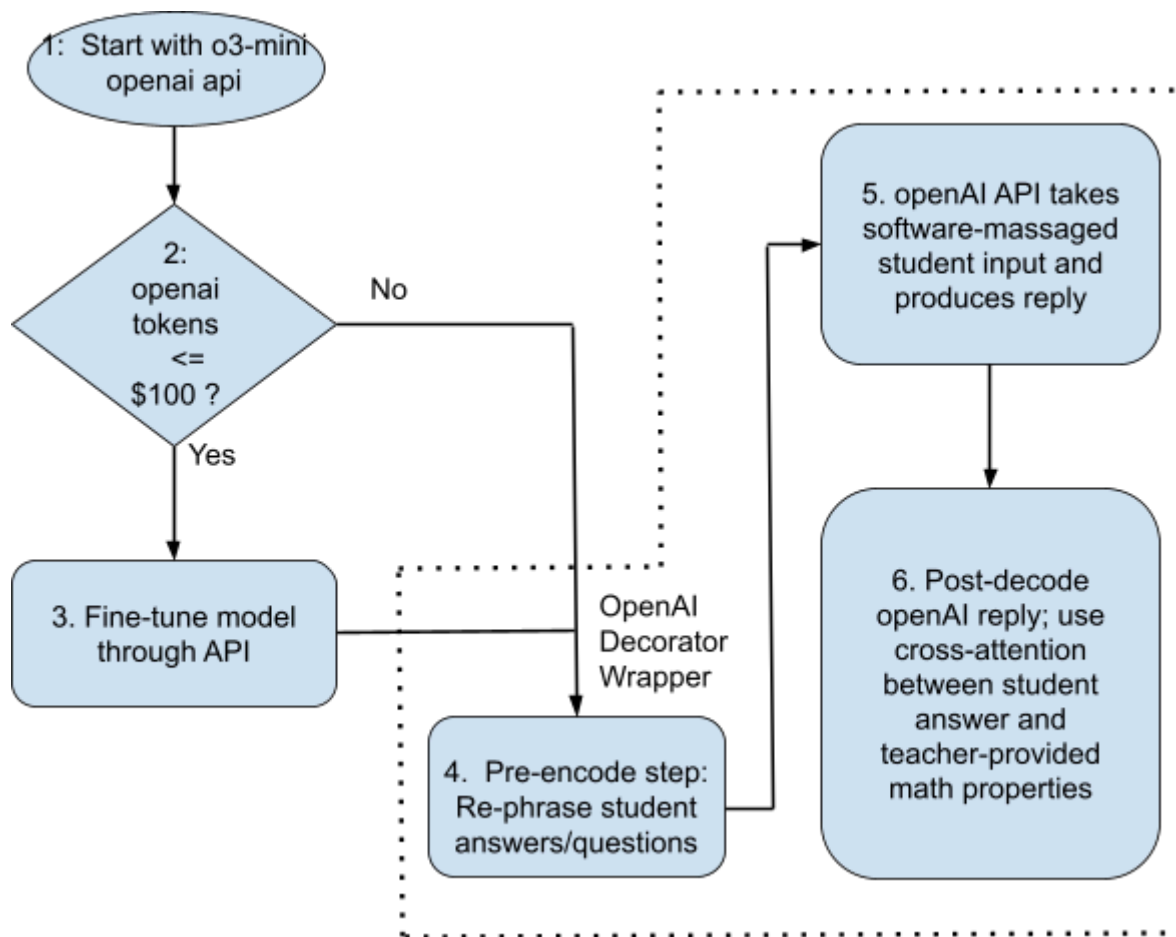


A Chatbot for Conducting Mental Arithmetic Conversations with Middle School Students

Create a natural language chatbot to converse with students in mental math exercises (math without writing). Software ingests number talk series created by human teachers, e.g.:

$4 * 25 = ?$
 $4 * 250 = ?$
 $1000 / 4 = ?$
 $996 / 4 = ?$

Software poses questions to students. Students answer but are required to provide verbal reasoning leading to conclusions (correct or incorrect answers don't matter that much), and software converses with students to help lead them to make their reasoning more productive in achieving correct answers over the long run, in particular asking productive smaller questions to students. Software is given correct answers to problems (by human teachers) math properties that exist (such as distributive, associative, commutative, maybe incorporated using a markup language), and some candidate usage of properties to get such answers.



The chatbot will converse with students in a manner that quantitatively increases student participation over time. The software is essentially learning (machine-learning) to conduct better math conversations, and helping human students to have better math conversations over time (human-learning).

The overall *design* of the chatbot will be to function as a decorator wrapper for the OpenAI API, pre-processing student natural-language inputs before passing them to OpenAI, and post-processing OpenAI natural-language responses before passing them to students.

3. (Point 3 in diagram above)

Fine-tune model through API using transcripts / human-modeled number talk conversations. A reasonably good training set for math conversations can be found at:

<https://github.com/eth-nlped/mathdial>

... however for this math chatbot project, training and even fine-tuning are not 100% necessary.

4. (Point 4 in diagram)

Pre-encode step: Re-phrase student answers/questions to combine student text with teacher-provided math properties (e.g. distributive, commutative, associative)

For example, suppose a student sees the problem " $996 \div 4 = ?$ " and gives an answer plus reasoning: "996 divided by 4 is equal to 6 because $9 + 9 + 6 = 24$ and $24 \div 4$ equals 6." (This is a wrong answer with wrong reasoning, but exactly the kind of situation that a math chatbot needs to freely *allow* and to give a good reply, leading to a productive conversation ultimately correcting the student's misconceptions.) Hypothetically the chatbot might be expected to productively reply, "That's a good attempt, <student's name> [chatbot must always give correct attribution to students' work whether correct or incorrect]. What is it about the number 996 that caused you to add $9 + 9 + 6$? Is $9 + 9 + 6$ the same as 996?"

In more detail, the job of the AI chatbot will be to input the incorrect human response to $996 / 4 = 6$ (wrong), and create an openAI query to be similar to: "A child incorrectly thinks that $996 / 4 = 6$ when correctly $996 / 4 = (1000 - 4) / 4 = (1000 + (-4)) * (1/4) = (1000 * (1/4)) + ((-4) * (1/4)) = (1000 / 4) + (-4/4) = 250 - 1 = 249$, using the distributive property. Ask a question that asks the child a productive question (not an answer) containing correct reasoning. The child might have incorrectly reasoned that $996 / 4 = 6$ because $9 + 9 + 6 = 24$ and $24 / 4 = 6$."

OpenAI will produce a blackbox response to the query.

5. The last job of the math chatbot will be to massage the OpenAI response before passing it to the student, using the chatbot's own decoder mechanism. The decoder shall use cross-attention mechanisms to post-process the OpenAI output and re-phrase the message in a manner learning to increase the time spent in voluntary student math participation, where the correctness of student responses also increases for other math problems over time.

The active responsibility of the chatbot is very limited... teachers will actually provide the math problems, as well as providing supporting facts that are meant to help students solve the problems ("scaffolding").

For example the example problem set (in this paper's introduction) could be expressed roughly in JSON:

```
[ ...
{ problem: "1000 / 4"
  answer: 250
  related_problems: [ '4 * 250' ]
},
{ problem: "996 / 4"
  answer: 249
  related_problems: [ '1000 / 4' ]
  related_properties: [ { property: 'distributive'
                        usage: '996 / 4 = (1000 - 4) / 4 = (1000 + (-4)) *
(1/4) = (1000 * (1/4)) + ((-4) * (1/4)) = (1000 / 4) + (-4/4) = 250 - 1 = 249'
                      }
                    ]
  common_errors: [ { error: '996 / 4 = 6'
                    Reasoning: '9 + 9 + 6 = 24, 24 / 4 = 6' }
                ]
}
]
```

The main ethical issue of using AI to facilitate arithmetic conversations with students is a threat to human jobs, that is to say teacher jobs. But good math teachers are in short supply and have been for a long time. With math talks (active math conversations without a need for strict and sometimes immediate right-or-wrong answers) there are few teachers well-trained to conduct the number-talk method, and AI agents capable of doing it could even do these conversations in different languages (Spanish/French/Chinese) using a language-translation wrapper, where there might be no trained teachers at all. This could be useful for students with English-as-Second-Language (ESL) needs. In the end, the math chatbot is meant to *assist* in teaching and to fill existing gaps in teaching, not to replace teachers.

The engineering design of this UMGC project will be to act as a productive mediator between the openAI API text-completion ability and students' answers/questions, so that the back-and-forth math conversation between students and math chatbot becomes ultimately more productive, more measurably productive. The score measuring productivity will be length-of-time that students voluntarily spend with the chatbot (the project is somewhat trying to create a "productive addiction" in a subject – math – a topic that most students dislike), and increased ability for students to be able to independently and correctly reason for other math problems-and-answers over the long-term.

Overall, the math number-talk chatbot is an attempt at incremental progress using AI for engaging students in more math conversations, and more productive math conversations.

History of Project Discussion:

Original proposal draft 4/27/2026:

[Idea 1 described below, selected...]

Natural language chatbot to converse with students about number talks (mental math exercises done by students without writing down calculations). Software ingests number talk series created by human teachers, e.g.:

$4 * 25 = ?$
 $4 * 250 = ?$
 $1000 / 4 = ?$
 $996 / 4 = ?$

Software poses questions to students. Students answer but are required to provide verbal reasoning leading to conclusion (correct or incorrect answer doesn't matter that much), and software converses with the student to help lead them to make their reasoning more productive in achieving correct answers over the long run, in particular asking productive smaller questions to students. Software converses with students in a manner that quantitatively increases student participation over time. The software is essentially learning to conduct better math conversations. Software is given (by human teachers) math properties that exist, and correct answers to problems and some candidate usage of properties (such as distributive, associative, commutative, maybe incorporated using a custom markup language) to get such answers.

Comments from Rajib 5/1/2026: Go ahead with idea 1. Need to find out specific use cases. What experience a student has and final objectives. Reasonable LLM with reasoning capability. Think of more details. Don't have to train model. Use GPT and have API calls.

List of Capstone project training dataset candidates 5/23/2026 (myang):

https://www.insidemathematics.org/sites/default/files/assets/classroom-videos/public-lessons/2nd-grade-math-addition-and-subtraction-word-problems-with-unknowns/transcript_number_talk_part_1.pdf

<https://github.com/eth-nlped/mathdial>

<https://huggingface.co/datasets/DeepMostInnovations/saas-sales-conversations>

[https://rajpurkar.github.io/SQuAD-explorer/explore/1.1/dev/Prime_number.html?model=r-net+%20\(ensemble\)%20\(Microsoft%20Research%20Asia\)&version=1.1](https://rajpurkar.github.io/SQuAD-explorer/explore/1.1/dev/Prime_number.html?model=r-net+%20(ensemble)%20(Microsoft%20Research%20Asia)&version=1.1)

Comment from Rajib 5/29/2026: Start Capstone project only after finishing Unit 24 (Optimization). Work on units 19, 20, 22, 24.

Thoughts before meeting 7/24/2026:

Transfer learning is the most important topic for the number talk chatbot. Use the training for other natural language chatbots, freeze that training, and add additional layers to customize for number talk conversations. Might not be able to re-train all layers afterwards with low learning rate (discuss with Rajib).

Comment from Rajib 7/24/2026:

No need to train model on additional number-talk data; just use API calls.